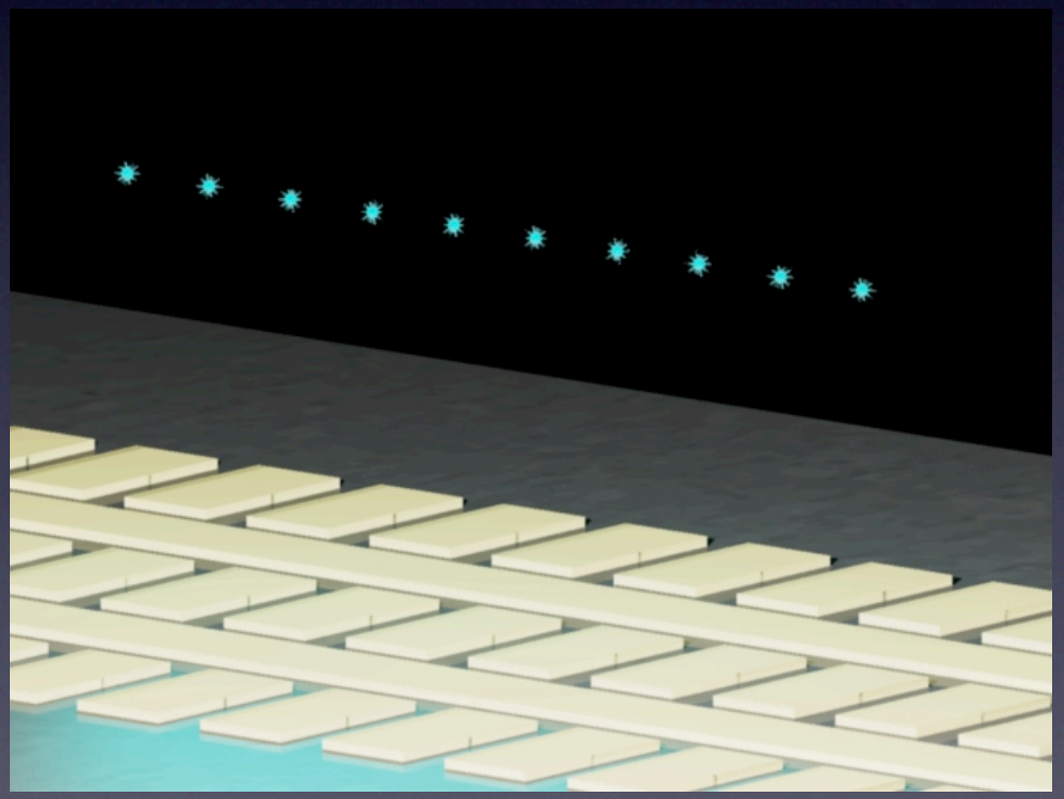
Building a "Large" Parallel Quantum Computer

Dave Bacon
University of Washington

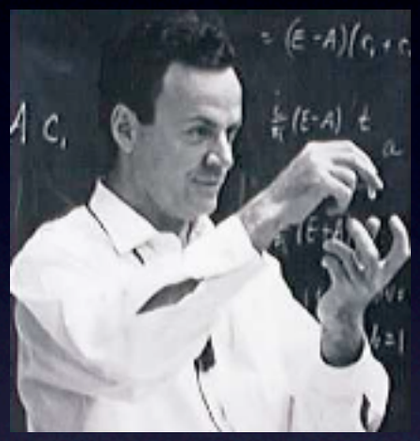


Funded by NSF and DARPA

“Parallel Processing For Scientific Computing”

- Parallelism in Quantum Computers
- Scientific Computing with Quantum Computers
- Can we build it? The MUSIQC Machine

Path To Quantum Computing



Richard Feynman



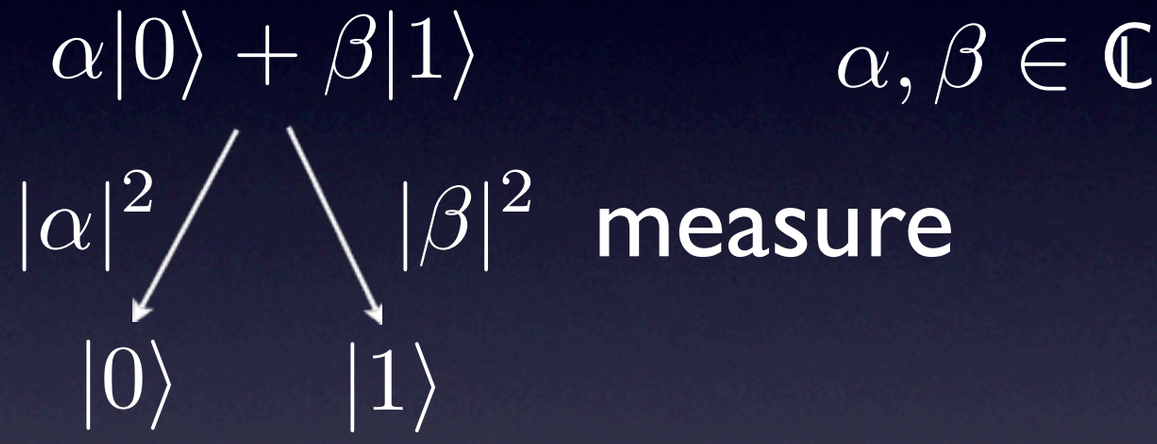
Dave Deutsch

1. Laws of physics are quantum mechanical
2. Quantum theory appears to be hard to simulate on our computers
3. Yet somehow nature "simulates"

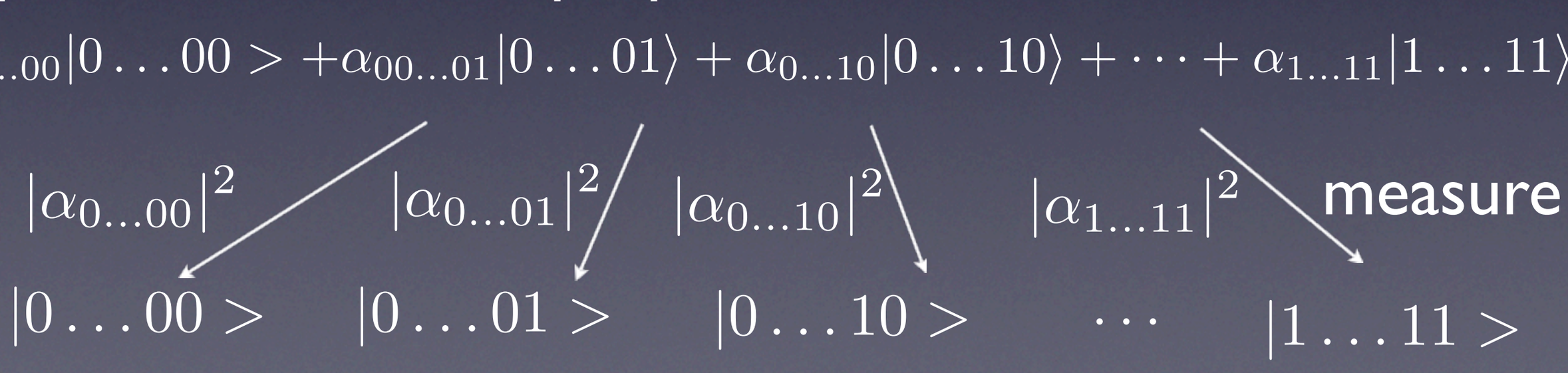
Computers where the information in the device is in its "quantum mechanical" form?

Qubit, Qubits

A single qubit can be in a "superposition" of the binary states 0 and 1:



n qubits can be in a superposition of 2^n states:



QC Parallelism Myth

Anyway, this is why quantum computing is so fast. “It's massively parallel computing,” X says. “Instead of doing things by steps, like in a computer program, it does everything at once.” - USA Today 2003

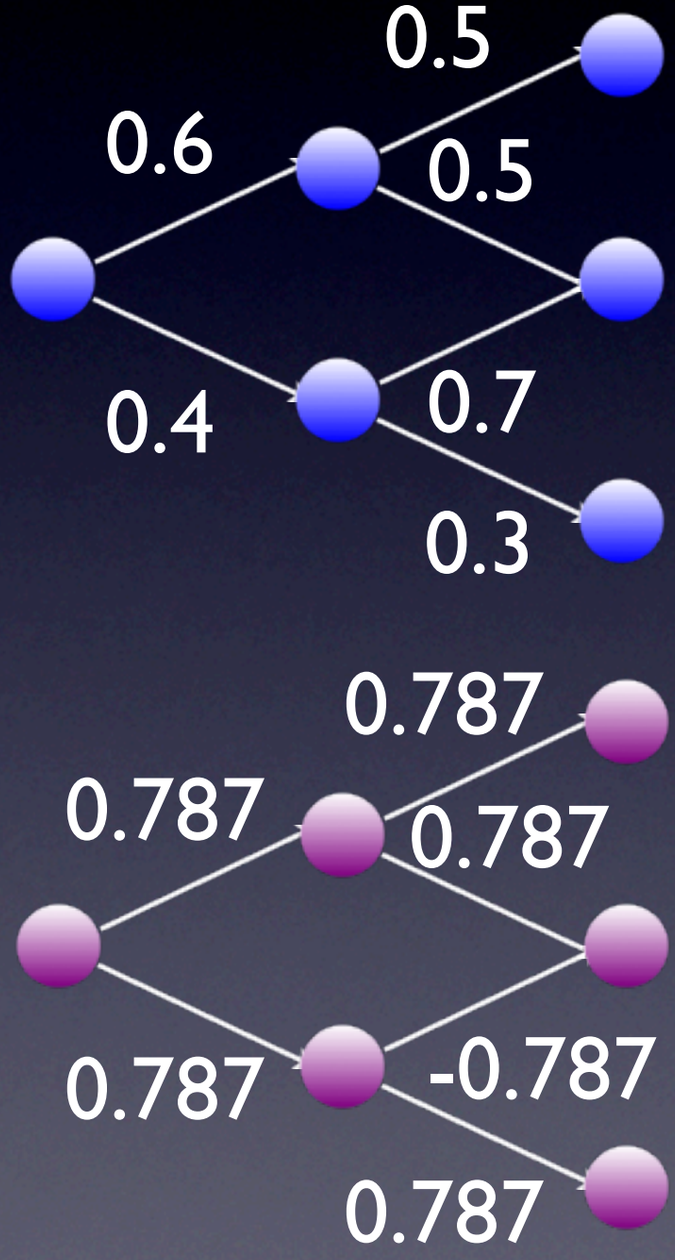
X = partner at well known VC firm

You can do 2^n simultaneous computation

$$\frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x, f(x)\rangle$$

but measure and you’ll see each with exponentially small probability!

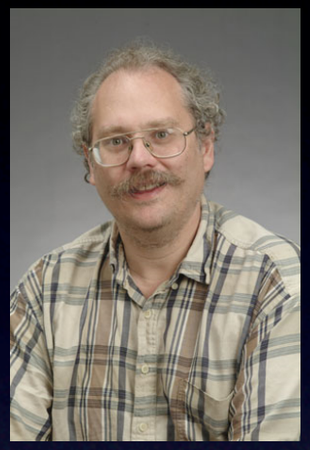
Garden of Re-forking Paths



$(0.6)(0.5)+(0.4)(0.7)=0.58$

$|(0.787)(0.787)+(-0.787)\ (0.787)|^2=0$

Shor's Breakthrough



Peter Shor

1881988129206079638386972394616504398071635633794173827007633564229888597152346654853190606065047430453173880113033967161996923212057340318795506569962213051687593076502570599807508642406493739712550055038649119906436234252670840638518957594638895726176858331794727721461074353025362230719730482246329146953020971164598521711305207112563635903975275

X

Best classical algorithm

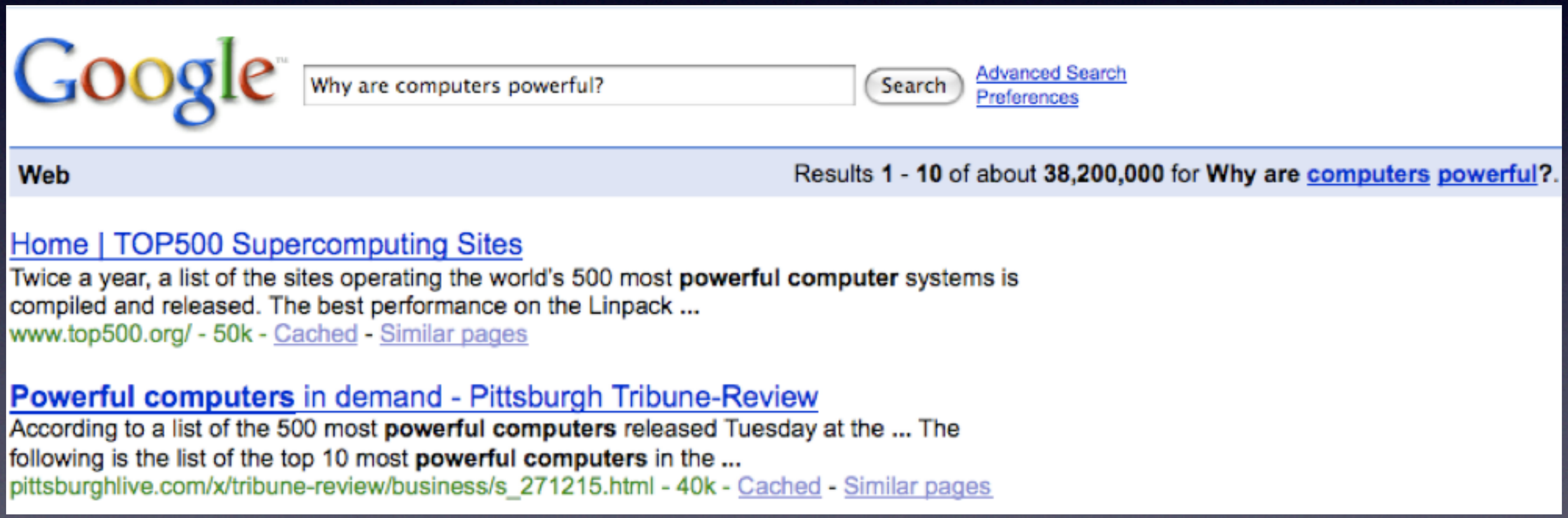
$$O\left(\exp(n^{\frac{1}{3}})\right)$$

Shor's quantum algorithm

$$O\left(n^3\right)$$

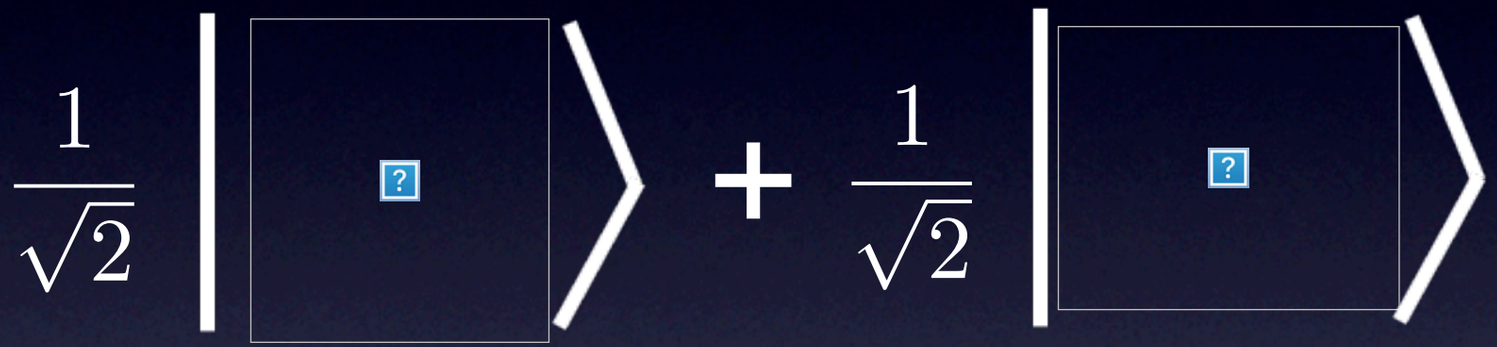
A Question

Where does the power of computers come from?



So Let's Build One!

Try to create a superposition

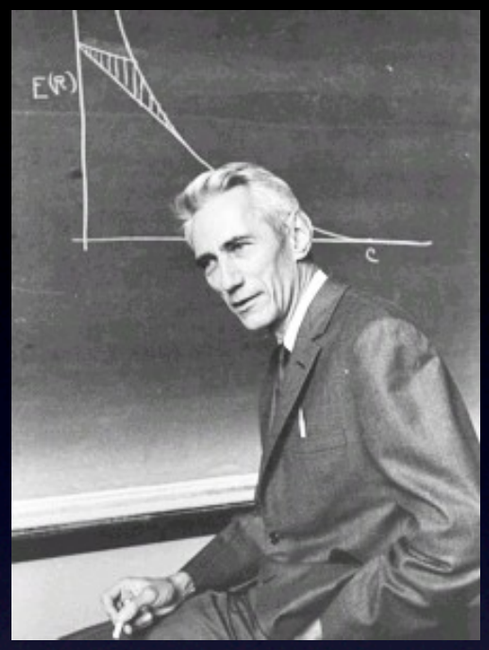


and it will quickly begin to behave classically



"decoherence"

Noisy Probabilistic Bits



Claude Shannon

How to communicate in the presence of noise?

"A mathematical theory of communication" (1948)

Error correcting codes



John von Neumann

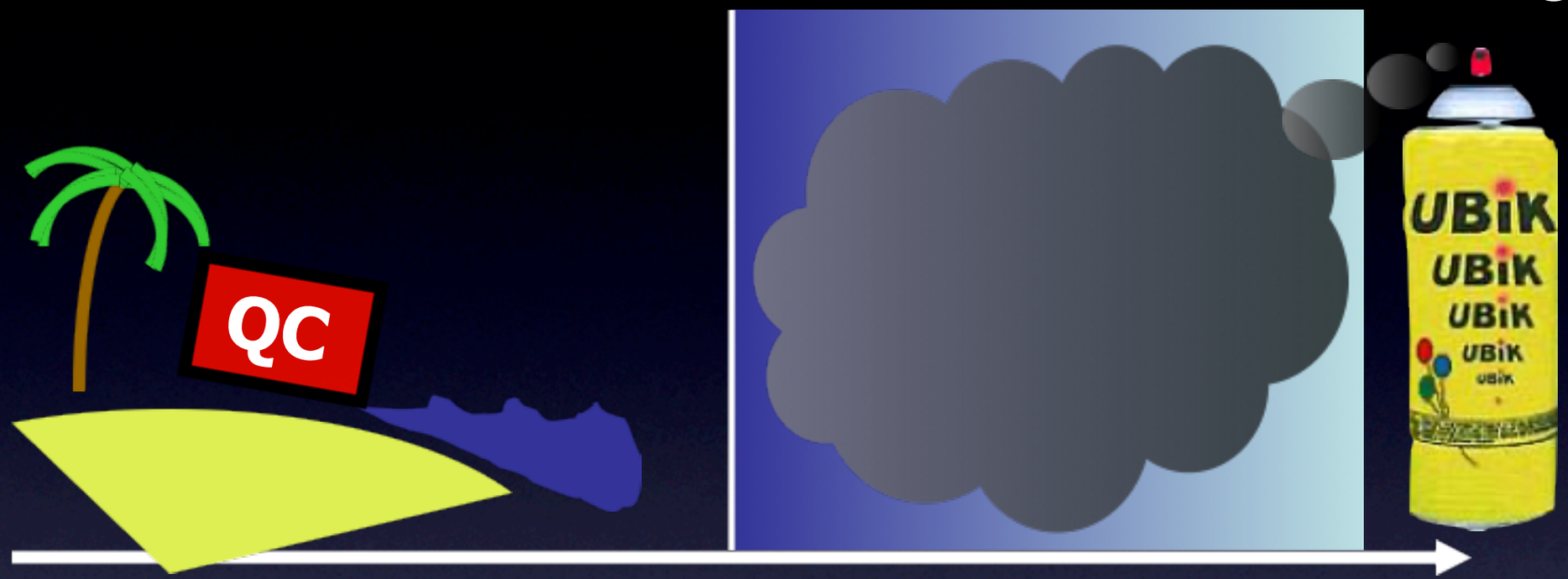
How to compute when the subsystems of your computer can fail?

"Probabilistic logics and the synthesis of reliable organisms from unreliable components" (1956)

Fault-tolerant architectures

Fault-Tolerance

errosol



“decoherence”

“imprecision”

Threshold Theorem for Fault-Tolerant Quantum Computation

[Aharonov, Ben-Or, Knill, LaFlamme, Zurek, Shor, Gottesman, Preskill]

If decoherence is weak enough and control is precise enough, then a quantum computation to a desired accuracy can be carried out with an overhead polynomial in the log of the accuracy.

QC Parallelism Fact

To build a large robust QC, one requires a parallel QC

un-decohered quantum information



some time later



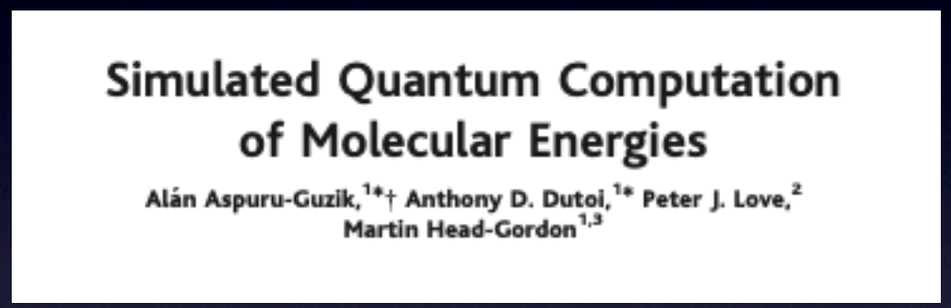
serial QC has kept coherent this quantum information

...but now the rest of the quantum information has decohered

Because decoherence is such a large problem, QC's must be highly parallel

Scientific QC

FactoriNg iS greAt if your a spook, but what if you're a scientist?


Simulated Quantum Computation of Molecular Energies

Alán Aspuru-Guzik,[1*†] Anthony D. Dutoi,[1*] Peter J. Love,[2] Martin Head-Gordon[1,3]

Science, 309, 1704 (2005)

~100 qubits one can compete with best classical molecular energy calculations



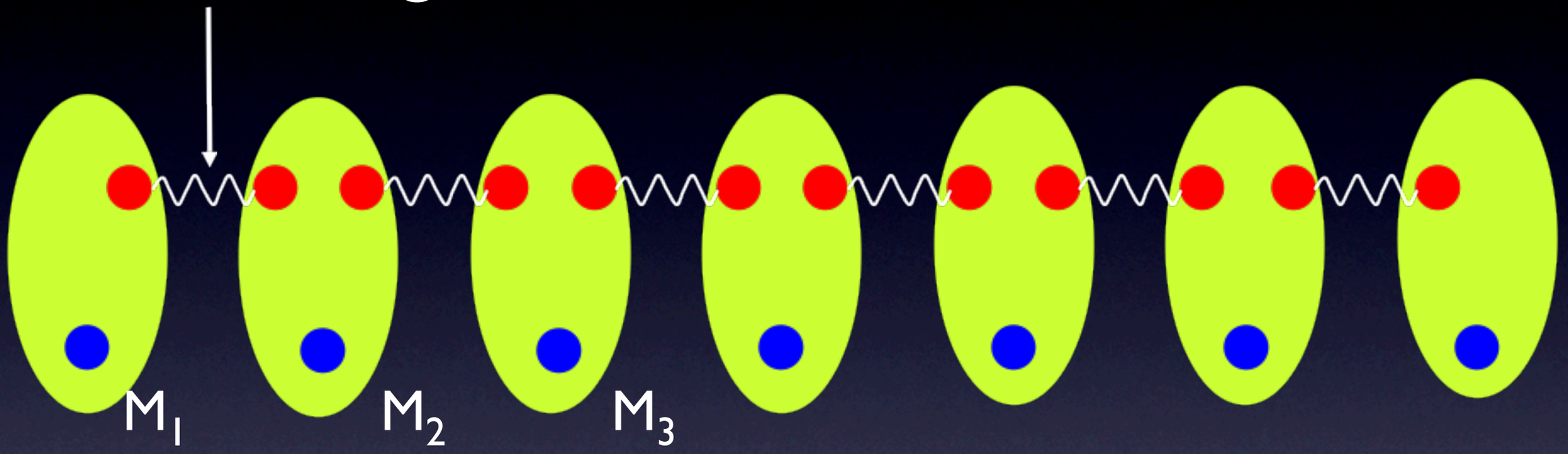
arXiv:0911.3635

QCs can overcome the "sign" problem of quantum monte carlo simulations

Annie Get Your Gun Principle

Scientific QC II

"Small" Entanglement = Efficient classical simulation



Density Matrix Renormalization Group
[White, 1991]

Quantum computer scientists have recently proven the convergence of DMRG

Quantum Hardware: Choices

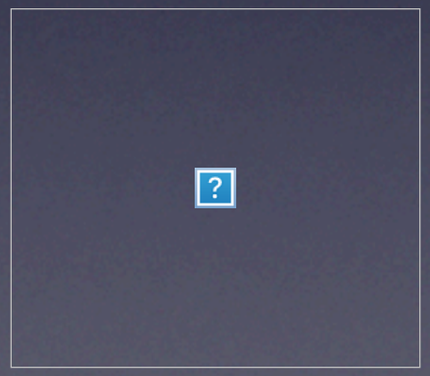
Solid State

superconducting circuits

electron spin in Phosphorus doped Silicon

quantum dots

defects in diamonds

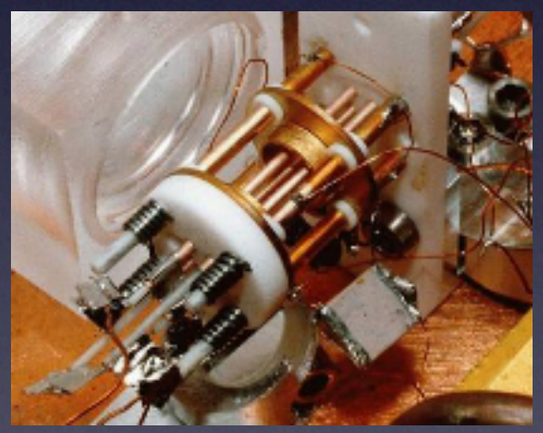


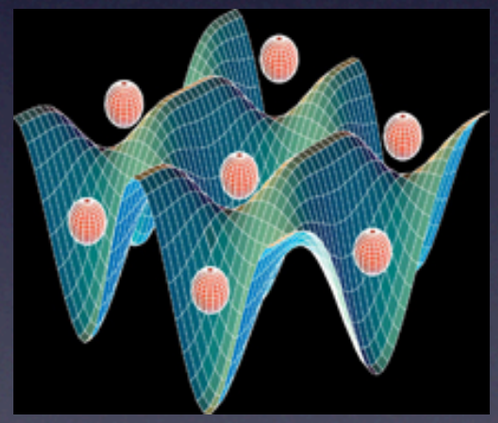
Atomic

cavity QED

neutral atoms in optical lattices

ion traps



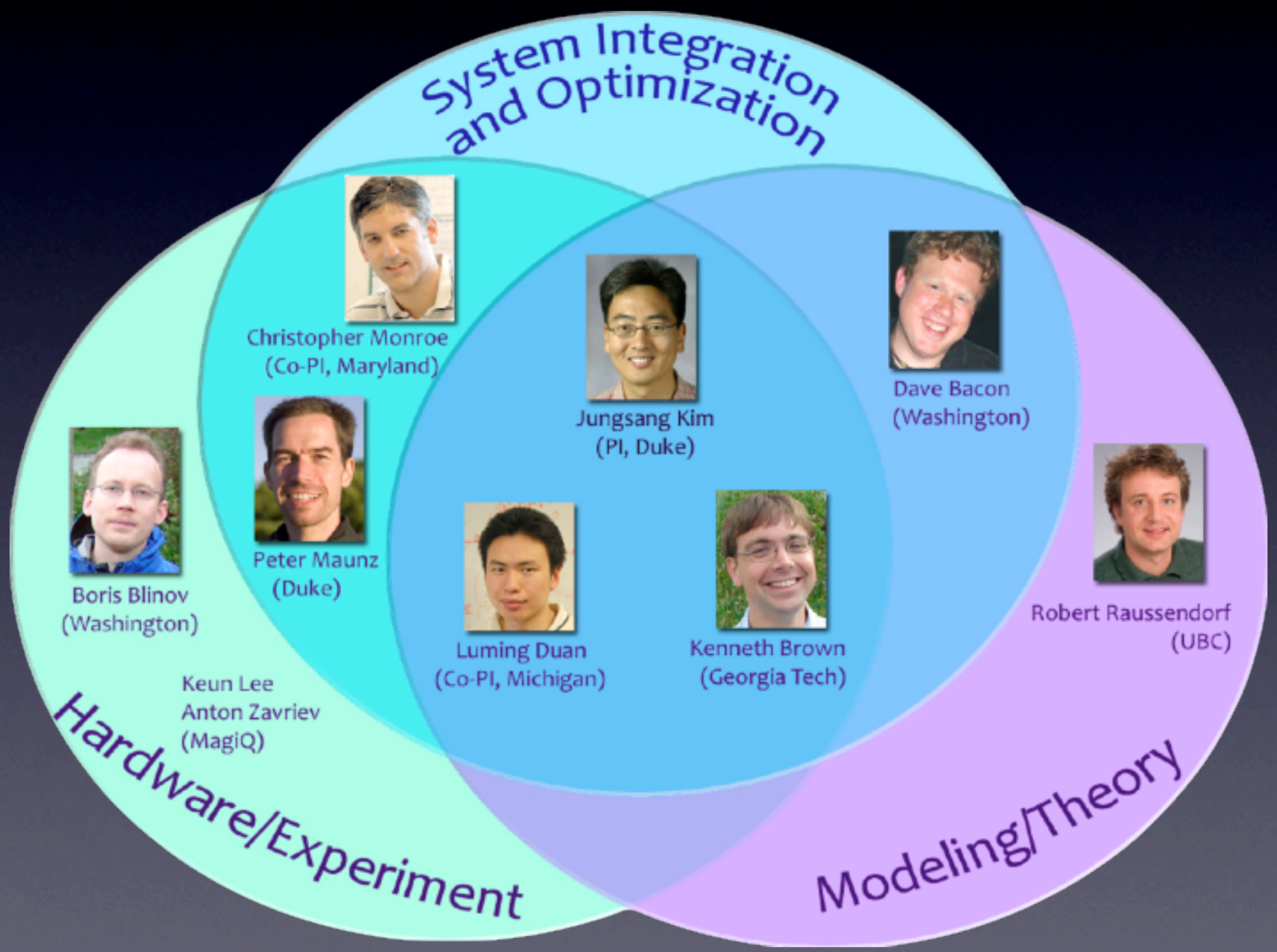


Photon Based

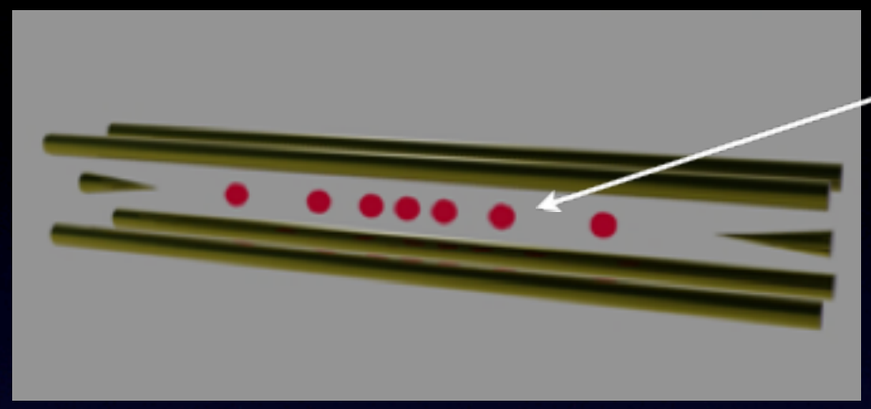
linear optics + single photon devices

MUSIQC

Modular Universal Scalable Ion-trap Quantum Computer



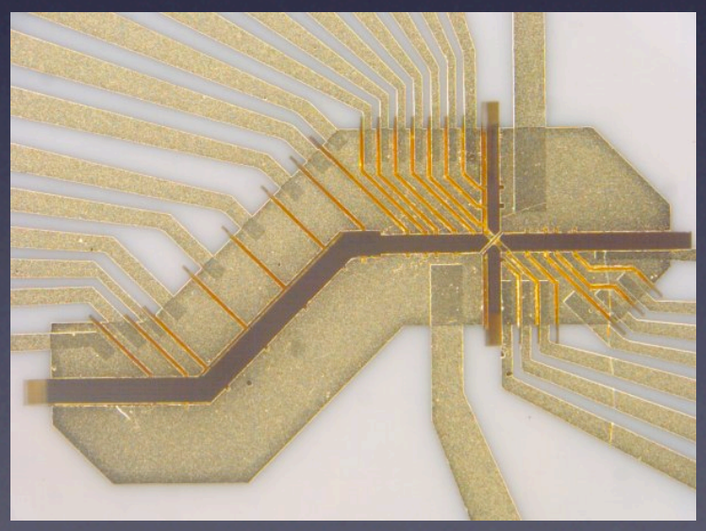
Ion Traps QC



$^{171}Yb^{+}$ ion = qubit

[$^{137}Ba^{+}$ / $^{40}Ca^{+}$]

Conventional (Paul) traps are limited in number of qubits they can store (linear crystal buckles and 2 qubit gate speeds slow)

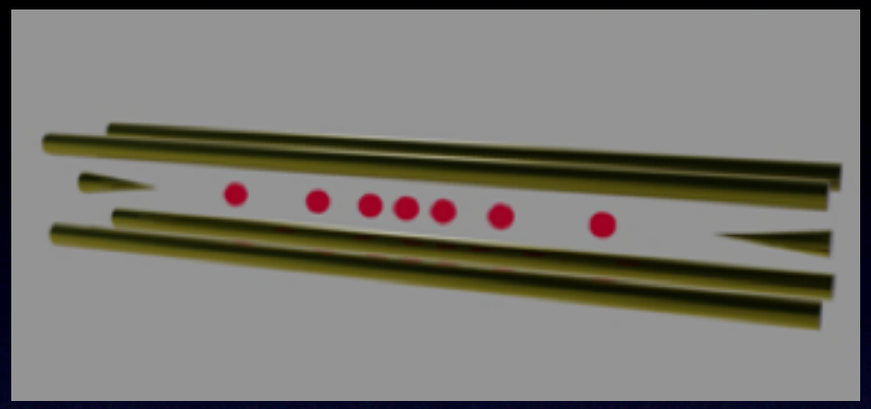


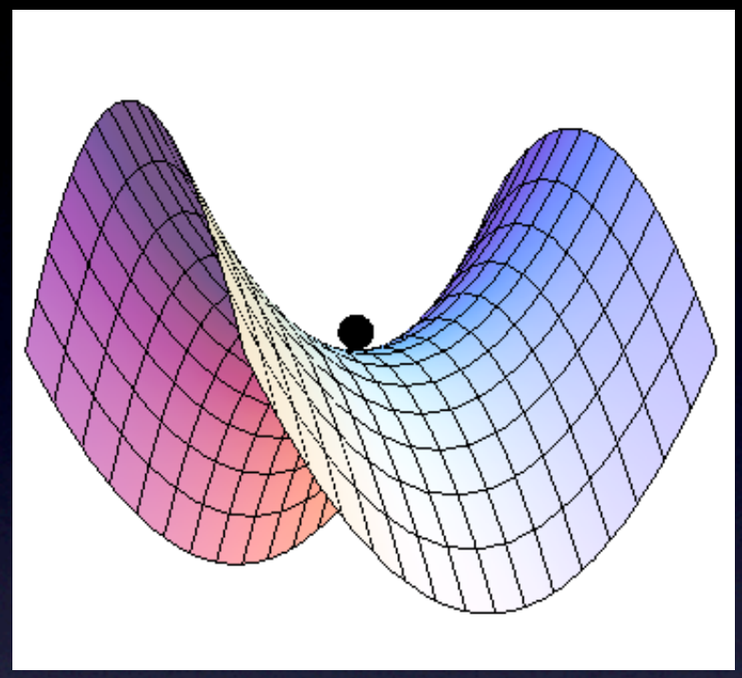
Physics Today (2009)
R.B. Blakestad (NIST)

Shuttle ions between trapping regions?

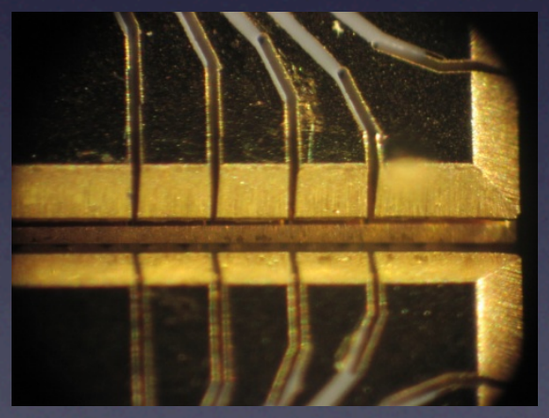
Problem: slow, heats up ion, technically challenging

Flat Traps



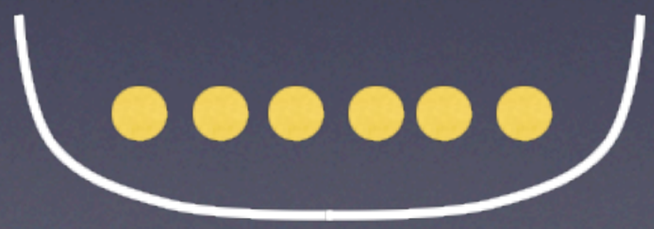


Harmonic confining potential along three directions



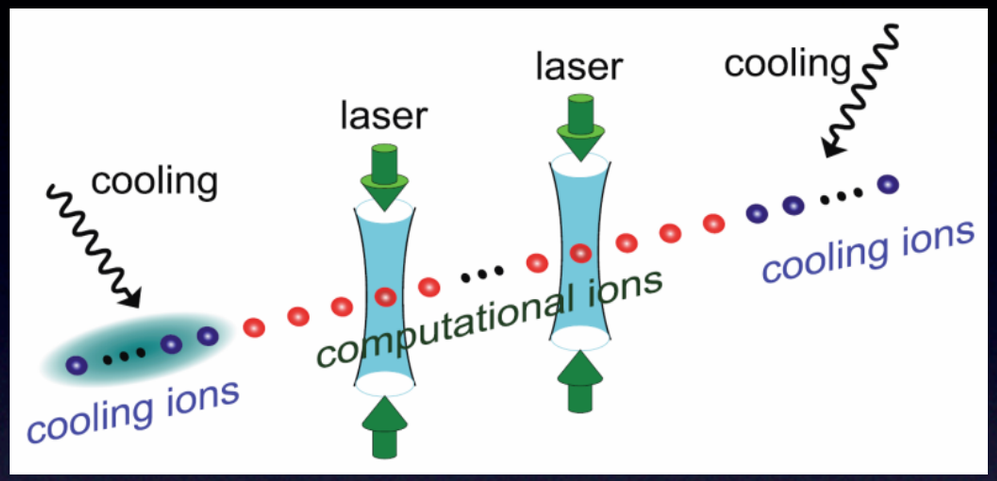
Au on Al_2O_3

But by adding voltage controls, one can build traps with "flat" bottoms



lots (>100, perhaps 1000) of nearly equally spaced ions

Flat Traps



Can no longer use longitudal mode to get ions to talk to each other

~10μm spacing

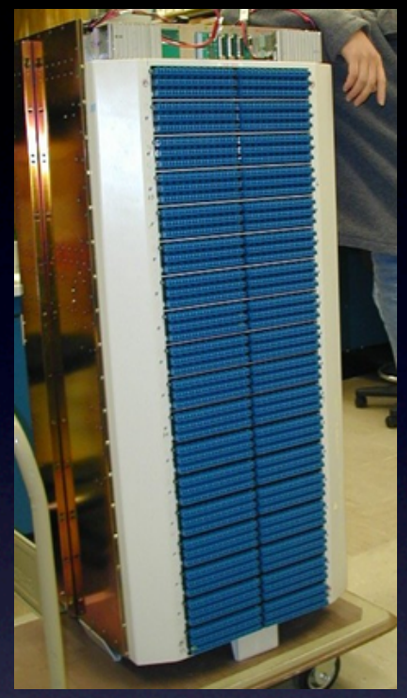
But by using transverse modes and pulse shaping, we can create multi-qubit gates between neighboring ions

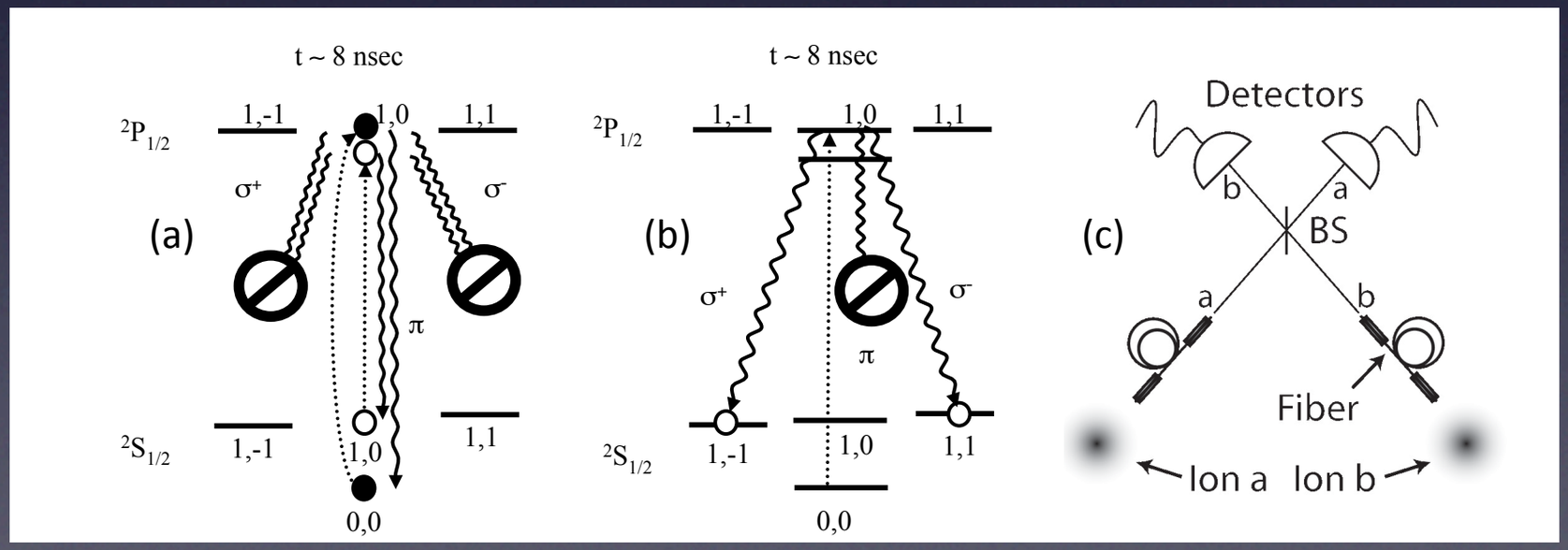
Phys.Rev.Lett.97,050505 (2006)

Addressing in Trap

- Use MEMS to address ions to trap
- Couple to cavity and cavity to fiber to communicate between traps

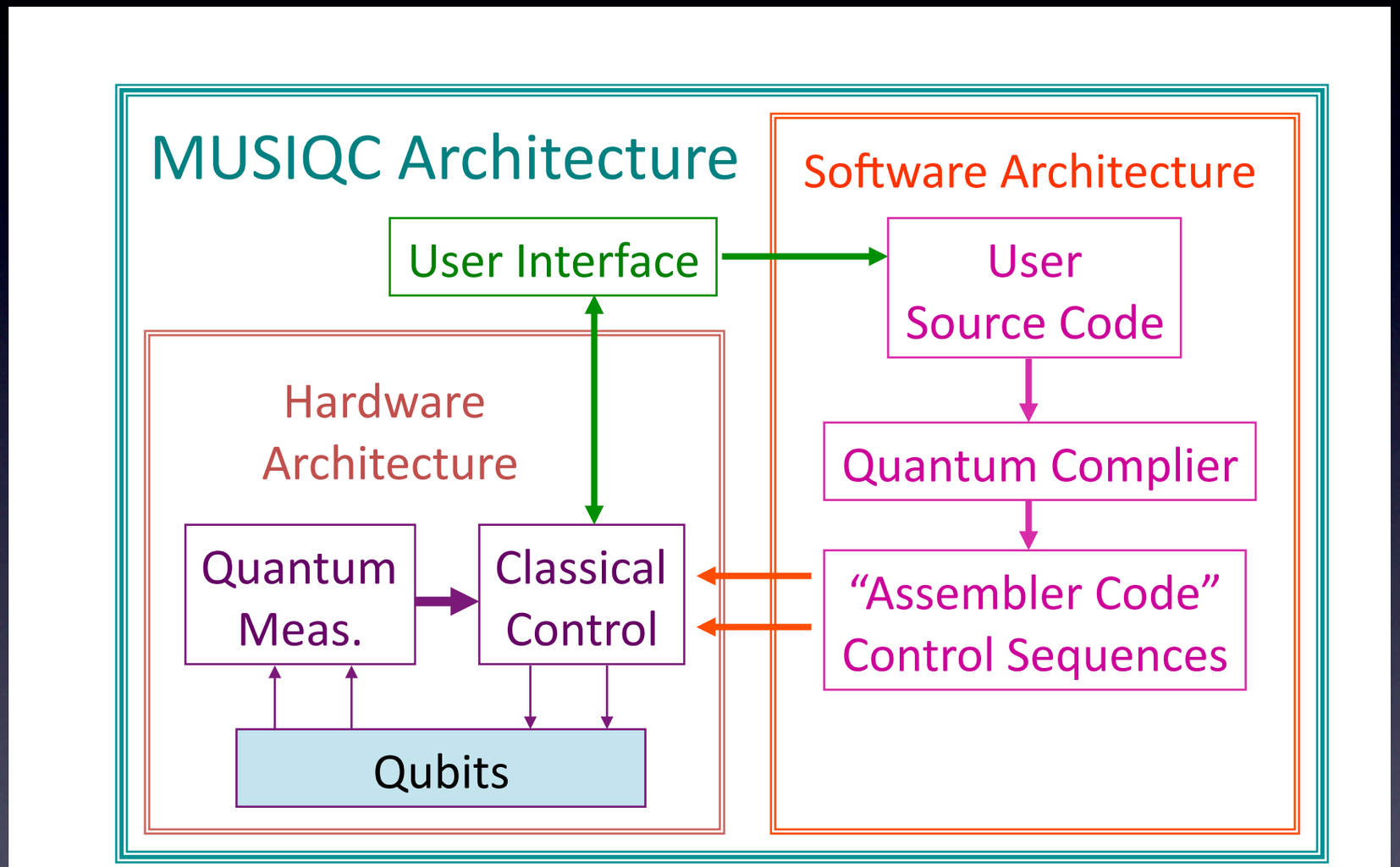
Connecting Up the Traps



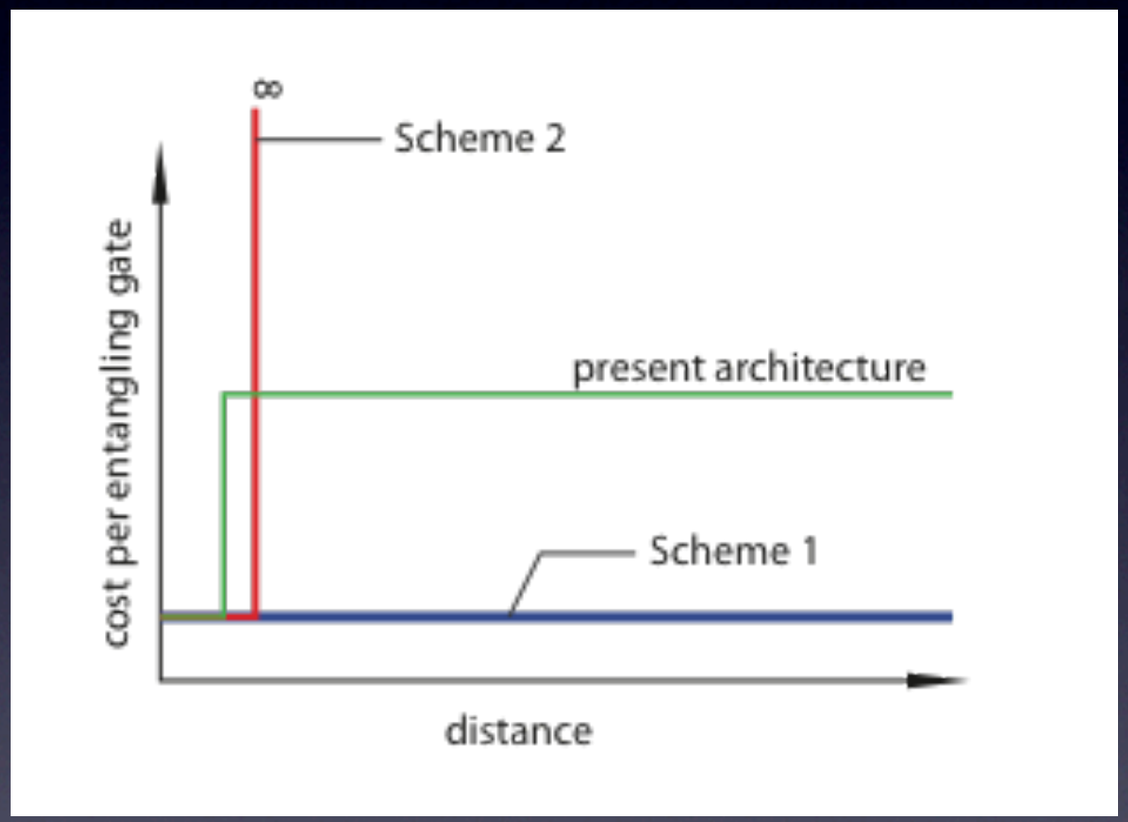


Probabilistic method for generating entanglement between traps

MUSIQC Architecture



A New Regime for Fault-Tolerance



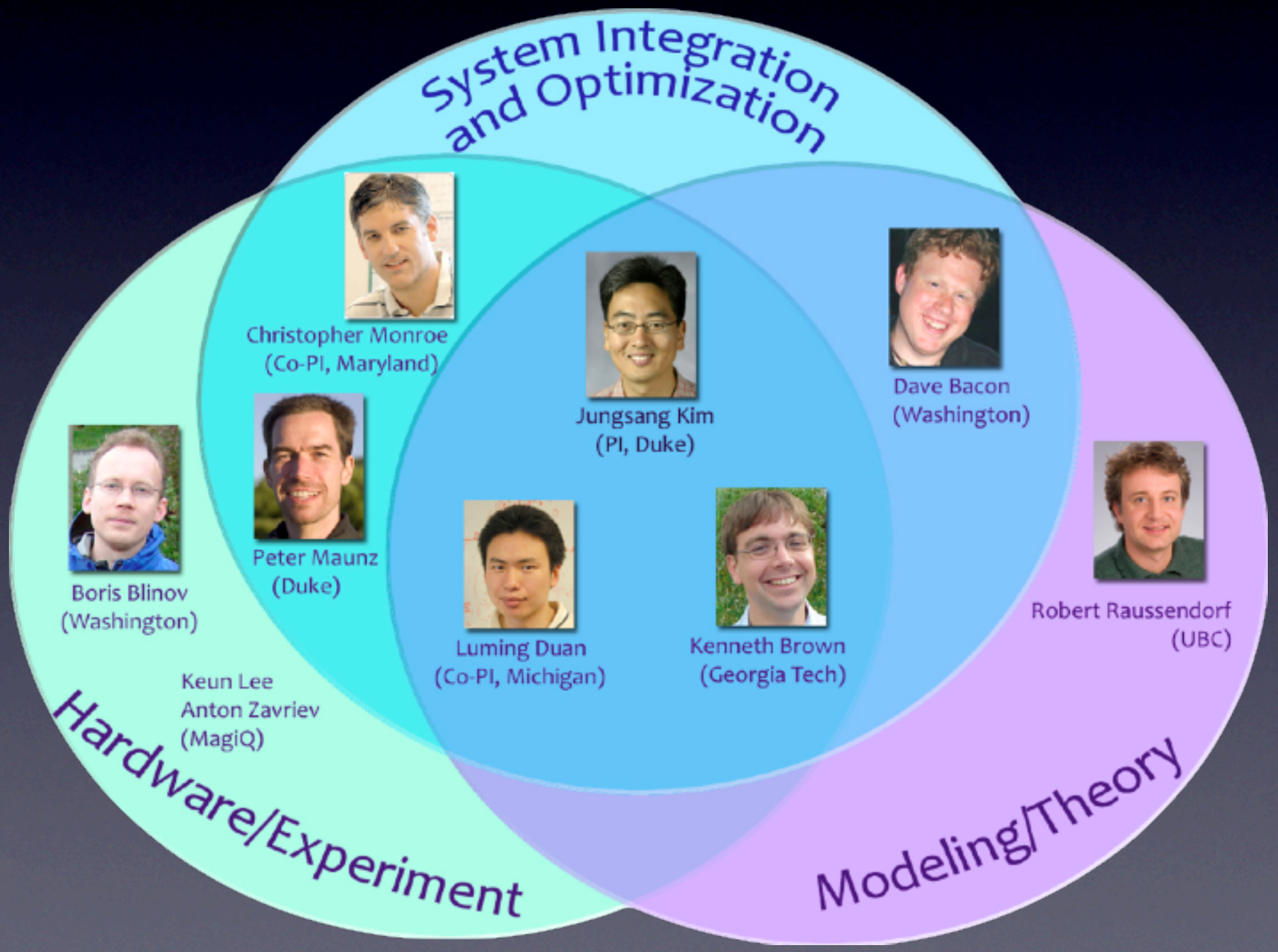
Scheme I = no cost for non-local gates
Scheme II = only spatially locally gates

How are the requirements for fault-tolerance shifted in this new architecture?

MUSIQC

- ~1.5 years to flat traps with 10s of qubit
- ~2 more years to connect 4 traps together
- Final result should be easily scalable to hundreds of qubits in hundreds of traps

MUSIQC - The Future of QC is Big Engineering



Quantum Computing Theory Group

Graduate Students

Gregory Crosswhite (Physics)
Isaac Crosson (Physics)
Lukas Svec (Physics)
Jijiang Yan (Physics)



funded by

NSF DARPA